3.12.1 The discovery of the electron (A-level only)

3.12.1.1 Cathode rays (A-level only)

Content
Production of cathode rays in a discharge tube.

3.12.1.2 Thermionic emission of electrons (A-level only)

Content
The principle of thermionic emission. Work done on an electron accelerated through a pd V ; $\frac{1}{2}mv^2 = eV$

Determination of the specific charge of an electron, $\frac{e}{m_e}$, by any one method.

Significance of Thomson's determination of $\frac{e}{m_e}$

Comparison with the specific charge of the hydrogen ion.

3.12.1.4 Principle of Millikan's determination of the electronic charge, *e* (A-level only)

Content
Condition for holding a charged oil droplet, of charge Q, stationary between oppositely charged parallel plates. $\frac{QV}{d} = mg$ Motion of a falling oil droplet with and without an electric field; terminal speed to determine the mass and the charge of the droplet. Stokes' Law for the viscous force on an oil droplet used to calculate the droplet radius. $F = 6\pi\eta rv$ Significance of Millikan's results. Quantisation of electric charge.